ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

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Project Name _____
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17.12CP Structural Checklist for Building Types C2: Concrete Shear Walls with Stiff Diaphragms and C2a: Concrete Shear Walls with Flexible Diaphragms

Table 17-24. Collapse Prevention Structural Checklist for Building Types C2 and C2a.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Low and Moderate Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CONCRETE BEARING WALLS: Floor and roof girders and trusses are not supported at the ends of concrete walls that are less than 10 in. (254 mm) thick. This statement only applies to framing supports located less than two times the wall thickness away from the wall end.	5.5.2.5.1	A.3.1.6.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: The number of lines of shear walls in each principal direction is greater than or equal to 2.	5.5.1.1	A.3.2.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SHEAR STRESS CHECK: The shear stress in the concrete shear walls, calculated using the Quick Check procedure of Section 4.4.3.3 is less than the greater of 100 lb/in.^{2} (0.69 MPa) or $2\sqrt{f'_c}$.	5.5.3.1.1	A.3.2.2.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REINFORCING STEEL: The ratio of reinforcing steel area to gross concrete area is not less than 0.0012 in the vertical direction and 0.0020 in the horizontal direction.	5.5.3.1.3	A.3.2.2.2	
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	WALL ANCHORAGE AT FLEXIBLE DIAPHRAGMS: Exterior concrete or masonry walls that are dependent on flexible diaphragms for lateral support are anchored for out-of-plane forces at each diaphragm level with steel anchors, reinforcing dowels, or straps that are developed into the diaphragm. Connections have strength to resist the connection force calculated in the Quick Check procedure of Section 4.4.3.7.	5.7.1.1	A.5.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	TRANSFER TO SHEAR WALLS: Diaphragms are connected for transfer of seismic forces to the shear walls.	5.7.2	A.5.2.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	FOUNDATION DOWELS: Wall reinforcement is doweled into the foundation with vertical bars equal in size and spacing to the vertical wall reinforcing directly above the foundation.	5.7.3.4	A.5.3.5	

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Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
High Seismicity (Complete the Following Items in Addition to the Items for Low and Moderate Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	DEFLECTION COMPATIBILITY: Secondary components have the shear capacity to develop the flexural strength of the components.	5.5.2.5.2	A.3.1.6.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	FLAT SLABS: Flat slabs or plates not part of the seismic-force-resisting system have continuous bottom steel through the column joints.	5.5.2.5.3	A.3.1.6.3	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COUPLING BEAMS: Coupling beams have stirrups spaced at or less than d/2, and each wall or wall segment connected to the coupling beam is supported such that it can resist shear and overturning forces in the absence of the coupling beam. This statement only applies to coupling beams with span-to-depth ratios exceeding 2-to-1.	5.5.3.2.1	A.3.2.2.3	
Diaphragms (Stiff or Flexible)							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	DIAPHRAGM CONTINUITY: Floor and roof diaphragms do not have expansion joints or vertical offsets, such as split levels, sawtooth, or clerestory configurations.	5.6.1.1	A.4.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	ROOF CHORD CONTINUITY: All chord elements are continuous, regardless of changes in roof elevation.	5.6.1.1	A.4.1.3	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	OPENINGS AT SHEAR WALLS: Diaphragm openings immediately adjacent to the shear walls are less than 25% of the wall length.	5.6.1.3	A.4.1.4	
Flexible Diaphragms							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CROSSTIES: There are continuous crossties between diaphragm chords.	5.6.1.2	A.4.1.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STRAIGHT SHEATHING: All straight-sheathed diaphragms have horizontal spans less than 24 ft (7.3 m) and aspect ratios less than 2-to-1 in the direction being considered.	5.6.2	A.4.2.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	DIAGONALLY SHEATHED AND UNBLOCKED DIAPHRAGMS: All diagonally sheathed or unblocked wood structural panel diaphragms have horizontal spans less than 40 ft (12.2 m) and aspect ratios less than or equal to 4-to-1.	5.6.2	A.4.2.2	

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C ☐	**NC** ☐	**N/A** ☐	**U** ☐	BLOCKED DIAPHRAGMS: All blocked wood structural panel diaphragms have horizontal spans less than 120 ft (36.5 m) and have aspect ratios less than or equal to 4-to-1.	5.6.2	A.4.2.3
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CANTILEVERED WOOD DIAPHRAGMS: All cantilevered wood diaphragms that provide lateral support for concrete or masonry walls consist of wood structural panels and have a maximum cantilever length of 20 ft (6.1 m) if unblocked or 35 ft (10.7 m) if blocked, and a maximum ratio of cantilever length to diaphragm width of 1:2 if unblocked and 1:1 if blocked. In addition, the cantilevered diaphragm has a back-span length equal to or greater than the cantilevered portion.	5.6.2	A.4.2.4
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	NON-CONCRETE-FILLED DIAPHRAGMS: Bare steel deck diaphragms or steel deck diaphragms with fill other than reinforced structural concrete consist of horizontal spans of less than 120 ft (36.5 m) and have aspect ratios less than 4-to-1.	5.6.3	A.4.3.1
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	OTHER DIAPHRAGMS: Diaphragms do not consist of a system other than wood, steel deck, concrete, or horizontal bracing.	5.6.5	A.4.7.1
Connections						
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	UPLIFT AT PILE CAPS: Pile caps have top reinforcement, and piles are anchored to the pile caps.	5.7.3.5	A.5.3.8

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